

Continuous functions – 5

The intermediate value theorem as a topological tool

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Theoretical minimum.

Theorem 1 (Bolzano and connectedness). A continuous image of a connected set is connected; on the line this is exactly the intermediate value property. The standard scheme: to find a point with a prescribed property, build an auxiliary continuous function whose zero is that point, and check that it changes sign.

Theorem 2 (Borsuk–Ulam on the circle). For every continuous $f: S^1 \rightarrow \mathbb{R}$ there is a pair of antipodal points with $f(x) = f(-x)$; equivalently, at any moment there are two diametrically opposite points of the equator with equal temperature.

Theorem 3 (Darboux). A derivative has the intermediate value property even when it is not continuous; hence the class of Darboux functions is strictly larger than the class of continuous ones.

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1. Prove Darboux's theorem, and then prove (RMO 1996) that for a differentiable f on an interval I the set $J = \left\{ \frac{f(b)-f(a)}{b-a} : a < b, a, b \in I \right\}$ is an interval, that $J \subseteq f'(I)$, and that $f'(I) \setminus J$ contains at most two points.
 2. Prove Theorem 2 and deduce the following: at every moment there are two antipodal points of the equator with the same temperature, and every continuous map of the circle to the line identifies at least one antipodal pair.
 3. (pancake theorem) Prove that any two bounded regions of the plane with positive area can be simultaneously bisected by a single straight

line. Then prove the one-dimensional ham sandwich statement: any bounded region can be bisected by a line in any prescribed direction.

4. (RMO) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ have the intermediate value property and be injective on $\mathbb{R} \setminus \mathbb{Q}$. Prove that f is continuous.
5. Prove that every continuous $f: [0, 1] \rightarrow [0, 1]$ has a fixed point, and deduce the one-dimensional Brouwer theorem for a closed disc: a continuous map of $[0, 1]^1$ into itself always has a fixed point, while for the open interval this fails.